

Class 7: Matrix for a Linear Transformation

Example: Prove that the function $T : \mathbf{R}^2 \rightarrow \mathbf{R}^3$ defined by $T(a, b) = (b, a + b, 5a)$ is a linear transformation.

T preserves vector addition:

T preserves scalar multiplication:

Key Idea: If T is any linear transformation from $\mathbf{R}^n$ to $\mathbf{R}^m$ and $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ are the columns of the $n \times n$ identity matrix, then the standard matrix for T is given by

Example: If $T(\vec{e}_1) = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$, $T(\vec{e}_2) = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ and $T(\vec{e}_3) = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix}$

what is the standard matrix for T ?

What is $T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right)$?

Key Idea: If T is any linear transformation and A is the standard matrix for T then $T(\mathbf{x}) = A\mathbf{x}$.

Two adjectives to describe linear transformations

A mapping T from $\mathbf{R}^n$ to $\mathbf{R}^m$ is said to be onto $\mathbf{R}^m$ if

A mapping T from $\mathbf{R}^n$ to $\mathbf{R}^m$ is said to be one-to-one if

Examples 1. $T(x, y, z) = (3x + 2y, x - 3y + z, y - z)$
What is the standard matrix for T ?

Find $\vec{x}$ so that $T(\vec{x}) = (1, 1, 1)$

$$2. T(\vec{x}) = A\vec{x} \text{ with } A \sim \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

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